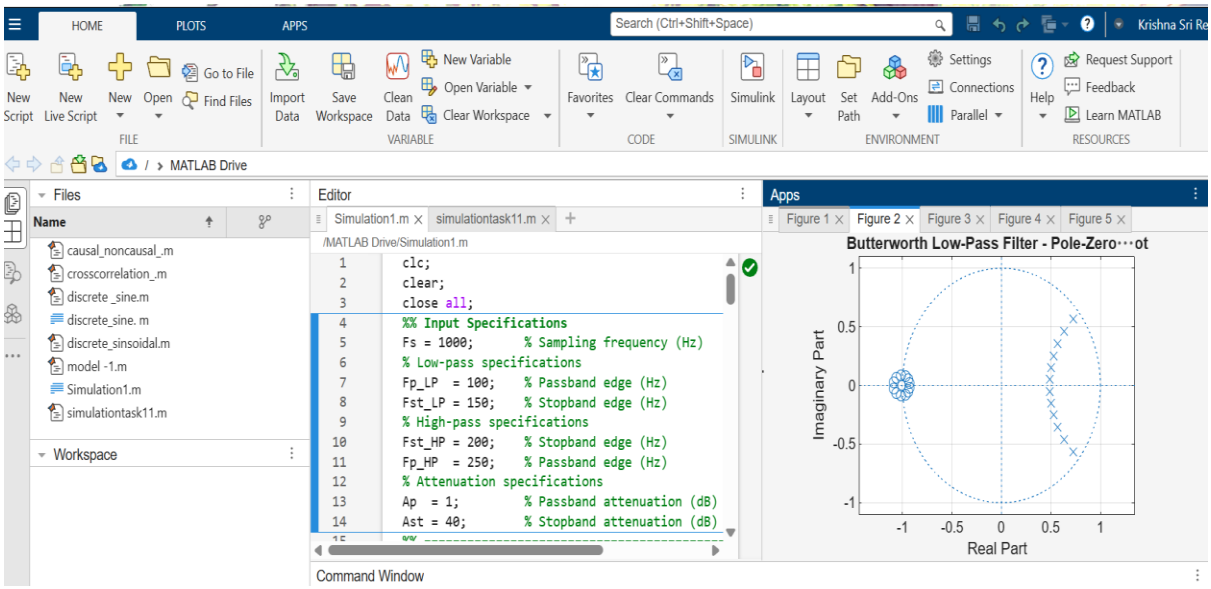
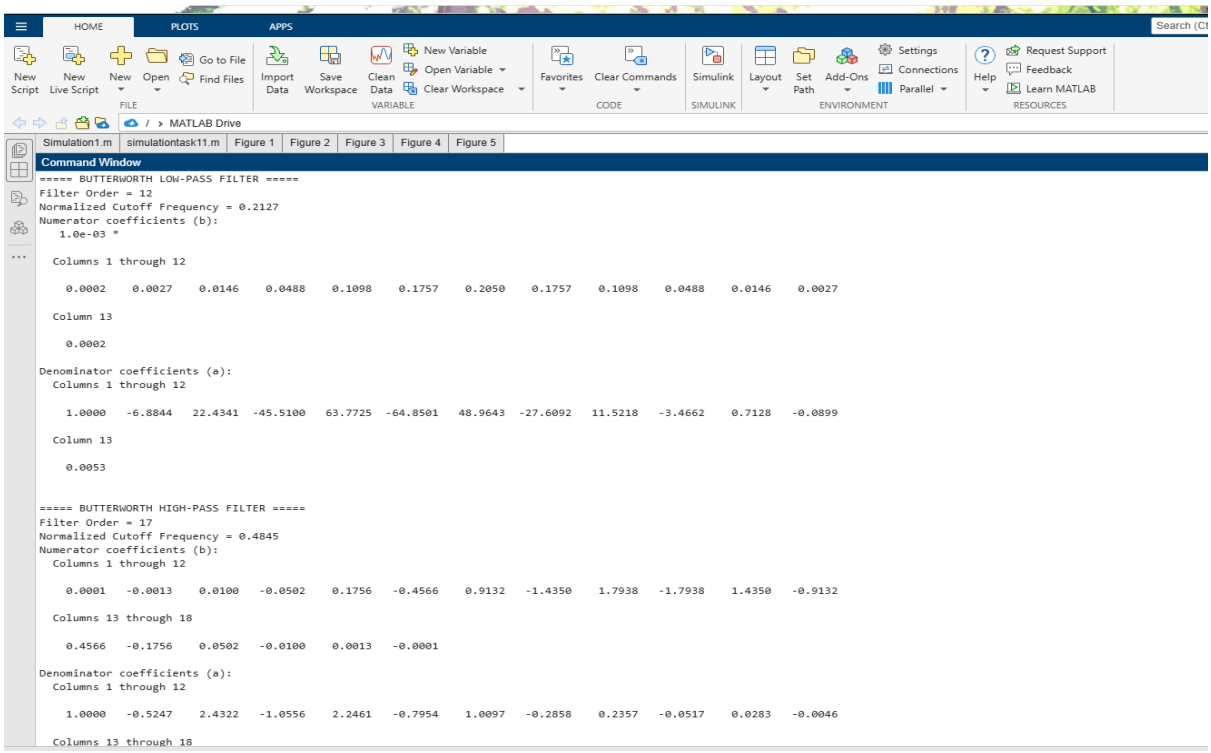

HOME
PLOTS
APPS
Search (Ctrl+Shift+Space)
Krishna Sri Re
New Script
New Live Script
New
Open
Go to File
Find Files
Import Data
Save Workspace
Clean Data
New Variable
Open Variable
Clear Workspace
Favorites
Clear Commands
Simulink
Layout
Set Path
Add-Ons
Settings
Connections
Parallel
Help
Request Support
Feedback
Learn MATLAB
FILE
VARIABLE
CODE
SIMULINK
ENVIRONMENT
RESOURCES
MATLAB Drive
Files
Name
causal_noncausal_m
crosscorrelation_m
discrete _sine.m
discrete_sine. m
discrete_sinsoidal.m
model -1.m
Simulation1.m
simulationtask11.m
Workspace
Editor
Simulation1.m
simulationtask11.m
/MATLAB Drive/Simulation1.m
1 clc;
2 clear;
3 close all;
4 %% Input Specifications
5 Fs = 1000; % Sampling frequency (Hz)
6 % Low-pass specifications
7 Fp_LP = 100; % Passband edge (Hz)
8 Fst_LP = 150; % Stopband edge (Hz)
9 % High-pass specifications
10 Fst_HP = 200; % Stopband edge (Hz)
11 Fp_HP = 250; % Passband edge (Hz)
12 % Attenuation specifications
13 Ap = 1; % Passband attenuation (dB)
14 Ast = 40; % Stopband attenuation (dB)
Apps
Figure 1
Figure 2
Figure 3
Figure 4
Figure 5
Butterworth Low-Pass Filter - Pole-Zero···ot
Imaginary Part
Real Part
1
0.5
0
-0.5
-1
-1
-0.5
0
0.5
1
Command Window


HOME
PLOTS
APPS
MATLAB Drive
Simulation1.m
simulationtask11.m
Figure 1
Figure 2
Figure 3
Figure 4
Figure 5
Command Window
===== BUTTERWORTH LOW-PASS FILTER =====
Filter Order = 12
Normalized Cutoff Frequency = 0.2127
Numerator coefficients (b):
1.0e-03 *
Columns 1 through 12
0.0002 0.0027 0.0146 0.0488 0.1098 0.1757 0.2050 0.1757 0.1098 0.0488 0.0146 0.0027
Column 13
0.0002
Denominator coefficients (a):
Columns 1 through 12
1.0000 -6.8844 22.4341 -45.5100 63.7725 -64.8501 48.9643 -27.6092 11.5218 -3.4662 0.7128 -0.0899
Column 13
0.0053
===== BUTTERWORTH HIGH-PASS FILTER =====
Filter Order = 17
Normalized Cutoff Frequency = 0.4845
Numerator coefficients (b):
Columns 1 through 12
0.0001 -0.0013 0.0100 -0.0502 0.1756 -0.4566 0.9132 -1.4350 1.7938 -1.7938 1.4350 -0.9132
Columns 13 through 18
0.4566 -0.1756 0.0502 -0.0100 0.0013 -0.0001
Denominator coefficients (a):
Columns 1 through 12
1.0000 -0.5247 2.4322 -1.0556 2.2461 -0.7954 1.0097 -0.2858 0.2357 -0.0517 0.0283 -0.0046
Columns 13 through 18